A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Soils: fertility, erosion & management - Set 1

Mark scheme

AQA spec 3.2.5; The physical environment; textbook pp. 158-169

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** B (about 40:40:20 sand:silt:clay) **1** Balanced mix gives good drainage and aeration from the sand fraction **1** Water retention and nutrient (ion) retention from the clay fraction – easy root penetration and cultivation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **loam** → a balanced mixture of sand, silt and clay	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) **1** Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand **1** With only 5% clay there is little surface to hold ions so nutrients are leached – low water retention too **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeability** → how easily water passes through / lets water through • **adsorb** → nutrient ions sticking to the surfaces of clay and humus • **leached** → nutrients or ions washed down out of the soil / washed away by water	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** C (clay) holds much water so has a high thermal capacity: warms and cools slowly – A warms rapidly after cold weather **1** Clay particles held together by capillary water so roots find it hard to push between them – penetration is easy in sandy A **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Shake a known mass of soil with water in a measuring cylinder and leave to settle – particles settle in order of size (sand first, then silt, then clay; organic matter floats) **1** Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	

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Question	Marking guidance	Marks
02.1	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2** Detritivores (worms, beetle larvae, millipedes, woodlice) physically break up dead organic matter, increasing its surface area, and release nutrients – worms create tunnels improving drainage and aeration **up to 2** Decomposers (bacteria, fungi) secrete enzymes that break down dead organic matter, releasing nutrient ions (ammonification) and forming humus which binds particles into crumb peds **up to 2** Nitrogen-fixing bacteria convert nitrogen gas to ammonium ions – nitrifying bacteria oxidise ammonium to nitrite then nitrate for root absorption **up to 2** Mycorrhizal fungi form symbiotic associations with roots increasing phosphate uptake **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ammonification** → ammonium released from dead organic matter and protein • **humus** → dark, decomposed organic matter in soil • **peds** → soil crumbs / aggregates of soil particles	6
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** This is the range of tolerance of most plants and soil biota – acidic soils leach nutrients and mobilise toxic aluminium, damaging root membranes **1** In alkaline soils phosphates become insoluble / unavailable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Interstitial spaces are reduced so aeration falls (aerobic processes such as nitrification and decomposition slow) and roots cannot penetrate **1** Infiltration falls so waterlogging is more likely and runoff / erosion increase **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Interstitial** → pore spaces / air spaces between the particles • **nitrification** → ammonium oxidised to nitrite and then nitrate • **Infiltration** → water soaking into the soil surface	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-161; AQA spec 3.2.5.1.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Soil loss increases as vegetation cover decreases: forest lowest (0.5), bare soil highest (60 t ha^{-1} yr^{-1}) - over 100 times more **1** Ploughing downslope loses nearly three times as much as contour ploughing for the same crop **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **contour ploughing** → ploughing along the slope rather than up and down it	2
03.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Leaves / leaf litter intercept raindrops so soil particles are not dislodged by rain splash **1** Roots bind soil particles together **1** Vegetation acts as a windbreak reducing wind velocity and the kinetic energy available to carry particles **1** Plants increase infiltration (root channels, organic matter) so less surface runoff **1** Organic matter / humus from vegetation binds particles into stable peds **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **windbreak** → a line of trees or hedge that slows the wind • **infiltration** → water soaking into the soil surface • **humus** → dark, decomposed organic matter in soil **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Furrows running downslope act as channels / gulleys that increase the velocity (kinetic energy) of runoff **1** Furrows along the contour trap the water so it loses energy, deposits particles and infiltrates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **contour ploughing** → ploughing along the slope rather than up and down it	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** In the dry season the soil (especially low-clay soils) is loose and dry with little cohesion so wind blows it away **1** In the wet season heavy rain causes rain splash and surface runoff when infiltration capacity is exceeded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface	2

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
04.1	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Overgrazing:** vegetation is eaten faster than it grows so soil is exposed – hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff **2** Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles – fewer worms so less aeration and lower infiltration capacity – fewer nutrients released so less vegetation cover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface • **humus** → dark, decomposed organic matter in soil **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Tree roots that held the soil together decompose so the soil is bound less strongly **1** Heavy rain adds mass and lubricates particles – without interception more water enters the soil, making slumping more likely **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Sedimentation in rivers reduces flow capacity so flooding downstream is more likely **1** Sedimentation in reservoirs reduces their storage volume **1** Coastal sedimentation covering reefs / seagrass – turbidity reducing light for aquatic plants **1** Increased atmospheric particulates from wind erosion worsening asthma – dust covering crops / urban areas **1** Desertification as vegetation cannot re-establish – reduced rainfall **1** Nutrients (phosphates) carried on soil particles causing eutrophication **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **particulates** → fine solid particles / smoke and dust • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
05.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 110 x 0.67 x 1.35 x (0.50 x 0.90) x 1.0 **1** = 44.8 t ha^{-1} yr^{-1} (accept 44.7-44.8) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1} yr^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 110 x 0.67 x 1.35 x (0.35 x 0.60) x 0.25 **1** = 5.2 t ha^{-1} yr^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1} yr^{-1}) is already printed on the answer line. • **contour ploughing** → ploughing along the slope rather than up and down it **Do not credit** • A bare wrong answer with no working.	2
05.3	**1 mark each** 1 mark for each of the 3 points below. **1 Beans / up-and-down ploughing:** severe class (over 30) and about 45 times the formation rate - clearly unsustainable, soil depth will fall **1** Wheat / mulch / strip cropping: moderate class, reduced by about 88% but still 5 times the formation rate so not fully sustainable **1** Further measures (long-term crops, terracing, windbreaks, more organic matter) or grazing land (C = 0.02) needed to bring loss below 1 t ha^{-1} yr^{-1} **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **terracing** → cutting the slope into level steps **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Factors are averages / arbitrary values for typical conditions – rainfall erosivity varies between years / storms **1** Does not include gully erosion, landslides, wind erosion or local variation in soil – based on data from particular regions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.5	1 mark for the point below. **1** C (crop / tillage) or P (erosion control practice) - R, K and LS are fixed by climate, soil and topography **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 167-169; AQA spec 3.2.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 5 points below. **1 Terracing:** sloping land divided into narrow level fields held by retaining walls along contours – water slows as it crosses each terrace and infiltrates instead of running off **1 Tied ridging:** on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff **1 Windbreaks:** hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles – protect bare fields after harvest **1 Mulching:** organic matter on the surface protects soil from raindrop impact and wind – decomposes to humus which binds particles – retains moisture **1 Strip cropping:** strips of different crops harvested at different times (at 90° to the wind) so there is always a crop protecting part of the field – slows runoff **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Terracing** → cutting the slope into level steps • **infiltration** → water soaking into the soil surface • **humus** → dark, decomposed organic matter in soil **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	5
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles **1** Permanent root systems and canopy cover protect the soil all year **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Control stocking density so vegetation regrows faster than it is eaten **1** Restrict access to vulnerable areas such as river banks (fencing) **1** Move / remove livestock at high-risk times, eg after heavy rain when soil is wet **1** Rotational grazing to rest pastures **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Water = 65.0 - 55.0 = 10.0 g – fresh soil = 40.0 g **1** 10.0 / 40.0 x 100 = 25.0% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Organic matter = 55.0 - 52.6 = 2.4 g – dry soil = 30.0 g **1** 2.4 / 30.0 x 100 = 8.0% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** So all water is removed and the mass loss on burning is due to organic matter only **1** Heating until mass no longer changes shows all water has evaporated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** Organic matter releases nutrients as it decomposes and provides food for soil biota **1** Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration **1** Surface organic matter / humus adhesion reduce rain splash and wind erosion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **peds** → soil crumbs / aggregates of soil particles	3

Do not write outside the box

Question	Marking guidance	Marks
07.5	1 mark for the point below. **1** Samples taken from different depths / locations (organic matter is concentrated near the surface) – incomplete burning – different heating times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - tiny pores between clay particles allow capillary rise – none in sand	1
08.2	**1 mark each** 1 mark for each of the 3 points below. **1 Hygroscopic:** held tightly by adhesive forces on particle surfaces – cannot drain and cannot be absorbed by plants **1 Capillary:** held in pores by surface tension – does not drain but can be absorbed by plants **1 Gravitational:** drains freely through large pores – available to plants only briefly as it passes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.3	1 mark for the point below. **1** Small round crumbs leave spaces giving good drainage, aeration and easy root penetration – large flat platy peds pack together reducing all three **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **peds** → soil crumbs / aggregates of soil particles	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	

Do not write outside the box